

TrackAlert Vision

Final Pin Diagram & Wiring Reference

Autonomous Indoor Patrol Robot · Arduino Mega 2560 + ESP32-S3-CAM

All VCC pins go to 5V and all GND pins go to the single common ground shared by the Arduino, motor-driver, buck converter and ESP32. Common ground is mandatory for the sensors and the board-to-board serial link to work.

1. Power architecture (two rails, one ground)

Rail	Details
Rail 1 - Logic	5V power bank -> Arduino Mega 5V and sensors
Rail 1 - Camera	5V power bank (USB-C) -> ESP32-S3-CAM
Rail 2 - Motors	4x18650 (approx 15V) -> ON/OFF switch -> LM2596 buck (set 6.5V) -> L298N +12V
L298N logic 5V	Arduino 5V -> L298N +5V pin (5V-EN jumper removed)
Common ground	Power bank GND + Arduino GND + buck OUT- + L298N GND + ESP32 GND all joined

2. L298N Motor Driver -> Arduino Mega

L298N pin	Arduino Mega	Note
ENA	D5	Left motor speed (PWM)
IN1	D44	Left direction
IN2	D45	Left direction
IN3	D46	Right direction
IN4	D47	Right direction
ENB	D6	Right motor speed (PWM)
+12V (VS)	Buck 6.5V	Motor supply
+5V	Arduino 5V	Logic (5V-EN jumper OFF)
GND	Common ground	Shared
OUT1 / OUT2	Left motor	
OUT3 / OUT4	Right motor	

Remove all three L298N jumper caps: the 5V-EN jumper (Arduino feeds 5V) and the ENA + ENB caps (so D5/D6 control speed).

3. 5-Array Line Following Sensor -> Arduino Mega

Sensor pin	Arduino Mega	Note
OUT1	D42	Digital
OUT2	D38	Digital
OUT3	D34	Digital
OUT4	D30	Digital
OUT5	D26	Digital
VCC	5V	Power
GND	Common ground	Shared

In code the array is read reversed: `LINE[5] = {26,30,34,38,42}`. Sensor reads 0 over the black line (`LINE_ON_BLACK = false`).

4. Servo + Ultrasonic (obstacle avoidance)

Component / pin	Arduino Mega	Note
SG90 Servo - Signal	D7	PWM (aims the ultrasonic)
SG90 Servo - VCC	5V	Power
SG90 Servo - GND	Common ground	Shared
HC-SR04 - TRIG	D28	Digital out
HC-SR04 - ECHO	D29	Digital in
HC-SR04 - VCC	5V	Power
HC-SR04 - GND	Common ground	Shared

5. Monitoring sensors

PIR (HC-SR501) - intruder

PIR pin	Arduino Mega	Note
Dout (OUT)	D32	HIGH on motion
Vcc	5V	Power
Gnd	Common ground	Jumper in H mode

Flame sensor - fire

Flame pin	Arduino Mega	Note
DO	D33	LOW = flame (digital)
VCC	5V	Power
GND	Common ground	Shared

MQ-2 Gas & Smoke

MQ-2 pin	Arduino Mega	Note
AO	A0	Analogue level
VCC	5V	Power (heater ~150mA)
GND	Common ground	Needs ~30s warm-up

6. I2C bus (shared SDA/SCL rail)

The MPU6050 and the 16x2 LCD share the same two I2C lines. Different addresses let them coexist.

Device	Arduino Mega	I2C address
MPU6050 - SDA	D20 (SDA)	0x68
MPU6050 - SCL	D21 (SCL)	0x68
MPU6050 - VCC / GND	5V / Common ground	
LCD 16x2 - SDA	D20 (SDA)	0x27
LCD 16x2 - SCL	D21 (SCL)	0x27
LCD 16x2 - VCC / GND	5V / Common ground	

7. Buzzer & Reset button

Component	Arduino Mega	Note
Active buzzer +	D8	Sounds on alert
Active buzzer -	Common ground	
Reset button - leg 1	D9	INPUT_PULLUP
Reset button - leg 2	Common ground	LOW when pressed - resumes patrol

8. Arduino Mega <-> ESP32-S3-CAM serial link

Connection	Detail	Note
Mega TX1 (D18)	-> 1k -> node -> ESP32 RX (GPIO21)	node -> 2k -> GND (5V to 3.3V divider)
Mega RX1 (D19)	<- ESP32 TX (GPIO47)	Direct, no divider
Mega GND	<-> ESP32 GND	Common ground (mandatory)

The Mega sends EVT:INTRUDER / EVT:FIRE / EVT:GAS on an alert; the ESP32 captures a photo and sends it to Telegram. The ESP32 is powered by USB-C from the power bank, but its GND must still join the common ground.

9. Complete Arduino Mega pin map

Mega pin	Connected to
D5	L298N ENA (left PWM)
D6	L298N ENB (right PWM)
D7	SG90 servo signal
D8	Buzzer (+)
D9	Reset / resume button (to GND)
D18 / TX1	-> 1k/2k divider -> ESP32 RX
D19 / RX1	<- ESP32 TX
D20 / SDA	MPU6050 SDA + LCD SDA
D21 / SCL	MPU6050 SCL + LCD SCL
D26	Line sensor OUT5
D28	HC-SR04 TRIG
D29	HC-SR04 ECHO
D30	Line sensor OUT4
D32	PIR Dout
D33	Flame DO
D34	Line sensor OUT3
D38	Line sensor OUT2
D42	Line sensor OUT1
D44 / D45	L298N IN1 / IN2
D46 / D47	L298N IN3 / IN4
A0	MQ-2 analogue out
5V	Sensors + L298N logic + ESP32 logic rail
GND	Common ground

Every pin is used once - no conflicts. This reflects the final assembled robot.

10. ESP32-S3-CAM key pins

ESP32-S3 pin	Function
5V / GND	Power from bank (USB-C) ; GND joins common ground
GPIO21	RX <- Mega TX1 via 1k/2k divider
GPIO47	TX -> Mega RX1
Camera / microSD / WiFi	Onboard (Freenove ESP32-S3-WROOM camera pinout)